

► PHARMACOLOGY—PHARMACOKINETICS AND PHARMACODYNAMICS

Enzyme kinetics

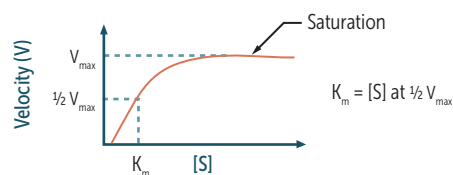
Michaelis-Menten kinetics

K_m is the substrate concentration needed for an enzyme to reach a rate of $1/2 V_{max}$ and is inversely related to the affinity of the enzyme for its substrate.

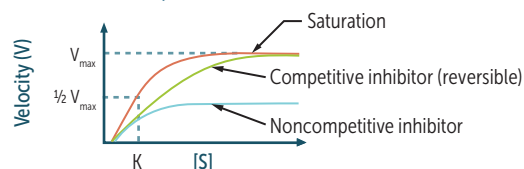
V_{max} is directly proportional to the enzyme concentration.

Most enzymatic reactions follow a hyperbolic curve (ie, Michaelis-Menten kinetics); however, enzymatic reactions that exhibit a sigmoidal curve usually indicate positive cooperativity in substrate binding (eg, aspartate transcarbamoylase).

$[S]$ = concentration of substrate; V = velocity.



Effects of enzyme inhibition



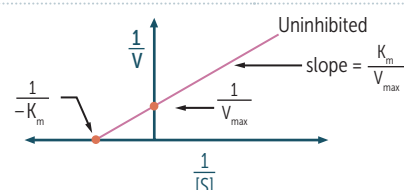
Lineweaver-Burk plot

The closer to 0 on the Y-axis, the higher the V_{max} .

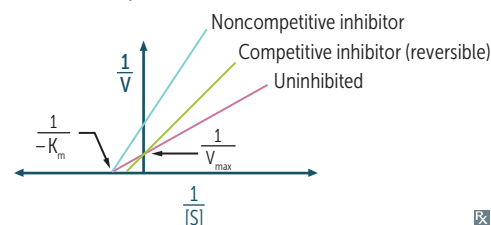
The closer to 0 on the X-axis, the higher the K_m .
The higher the K_m , the lower the affinity.

Reversible **competitive** inhibitors **cross** each other, whereas **noncompetitive** inhibitors do **not**.

Competitive inhibitors **increase** K_m .



Effects of enzyme inhibition

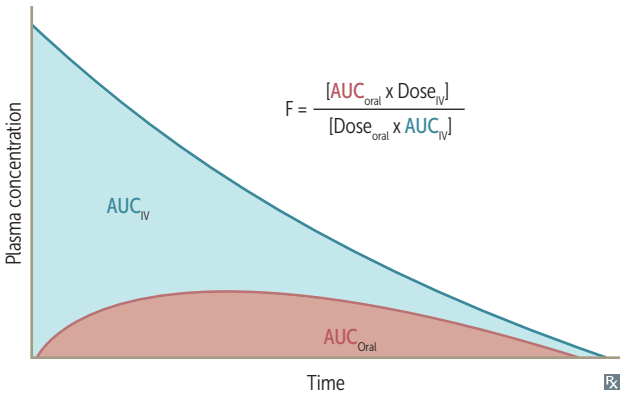


	Competitive inhibitors, reversible	Competitive inhibitors, irreversible	Noncompetitive inhibitors
Resemble substrate	Yes	Yes	No
Overcome by ↑ [S]	Yes	No	No
Bind active site	Yes	Yes	No
Effect on V_{max}	Unchanged	↓	↓
Effect on K_m	↑	Unchanged	Unchanged
Pharmacodynamics	↓ potency	↓ efficacy	↓ efficacy

Pharmacokinetics

Bioavailability (F)

Fraction of administered drug reaching systemic circulation. For an IV dose, F = 100%. Orally: F typically < 100% due to incomplete absorption and first-pass metabolism. Can be calculated from the area under the curve in a plot of plasma concentration over time.



Volume of distribution (V_d)

Theoretical value that relates drug amount to plasma concentration. Liver and kidney disease increase V_d (↓ protein binding, ↑ V_d). Drugs may be distributed in more than one compartment. Hemodialysis is most effective for drugs with a low V_d.

$$V_d = \frac{\text{amount of drug in the body}}{\text{plasma drug concentration}}$$

V _d	COMPARTMENT	DRUG TYPES
Low	Intravascular	Large/charged molecules; plasma protein bound
Medium	ECF	Small hydrophilic molecules
High	All tissues including fat	Small lipophilic molecules, especially if bound to tissue protein

Clearance (CL)

The volume of plasma cleared of drug per unit time. Clearance may be impaired with defects in cardiac, hepatic, or renal function.

$$CL = \frac{\text{rate of elimination of drug}}{\text{plasma drug concentration}} = V_d \times K_e \text{ (elimination constant)}$$

Half-life (t_{1/2})

The time required to eliminate 1/2 of the drug from the body. Steady state is a dynamic equilibrium in which drug concentration stays constant (ie, rate of drug elimination = rate of drug administration). In first-order kinetics, a drug infused at a constant rate takes 4–5 half-lives to reach steady state. It takes 3.3 half-lives to reach 90% of the steady-state level.

$$t_{1/2} = \frac{0.7 \times V_d}{CL} \text{ in first-order elimination}$$

# of half-lives	1	2	3	4
% remaining	50%	25%	12.5%	6.25%

Dosage calculations

$$\text{Loading dose} = \frac{C_p \times V_d}{F}$$

$$\text{Maintenance dose} = \frac{C_p \times CL \times \tau}{F}$$

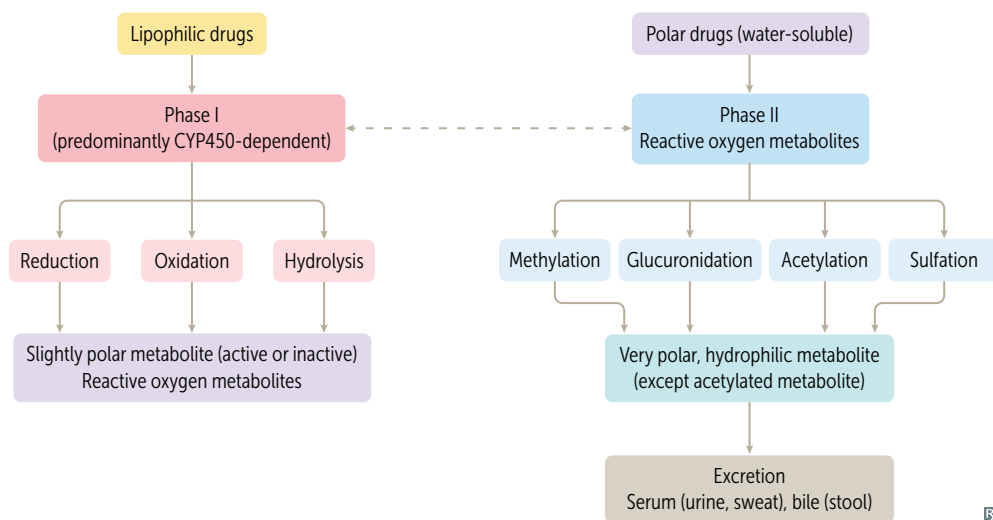
C_p = target plasma concentration
τ = dosage interval (time between doses); does not apply for continuous infusions

In renal or liver disease, maintenance dose ↓ and loading dose is usually unchanged. Time to steady state depends primarily on t_{1/2} and is independent of dose and dosing frequency.

Drug metabolism

Drugs can be metabolized by either or both phase I and phase II reactions. These reactions serve to bioactivate or deactivate substances, and do not have to take place sequentially (eg, phase I can follow phase II, or take place as a single reaction).

Geriatric patients lose phase I first. Patients who are slow acetylators have ↑ adverse effects from certain drugs because of ↓ rate of metabolism (eg, isoniazid).



Elimination of drugs

Zero-order elimination

Rate of elimination is constant regardless of C_p (ie, constant **amount** of drug eliminated per unit time). C_p ↓ linearly with time. Examples of drugs—**P**henytoin, **E**thanol, and **A**spirin (at high or toxic concentrations).

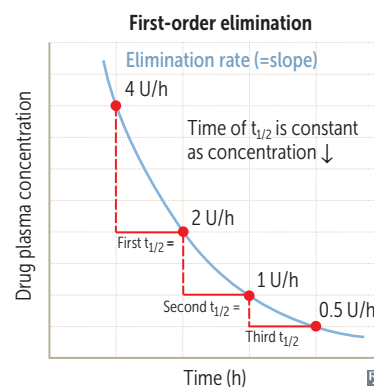
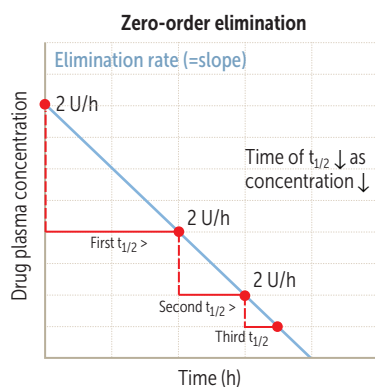
Capacity-limited elimination.

A **PEA** is round, shaped like the “0” in **zero**-order.

First-order elimination

Rate of **first-order** elimination is directly proportional to the drug concentration (ie, constant **fraction** of drug eliminated per unit time). C_p ↓ exponentially with time. Applies to most drugs.

Flow-dependent elimination.

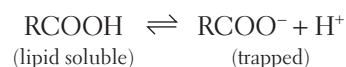


Urine pH and drug elimination

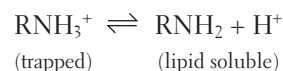
Ionized species are trapped in urine and cleared quickly. Neutral forms can be reabsorbed.

Weak acids

Examples: phenobarbital, methotrexate, aspirin (salicylates). Treat overdose with sodium bicarbonate to alkalinize urine.

**Weak bases**

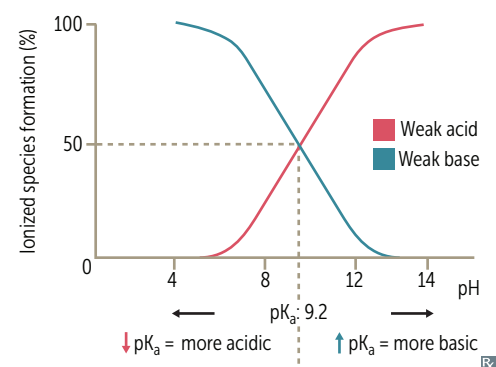
Examples: tricyclic antidepressants (TCAs), amphetamines. Trapped in acidic environments. For severe alkalosis, treat with ammonium chloride to acidify urine.



TCA toxicity is initially treated with sodium bicarbonate to overcome the sodium channel-blocking activity of TCAs. This treats cardiac toxicity, but does not accelerate drug elimination.

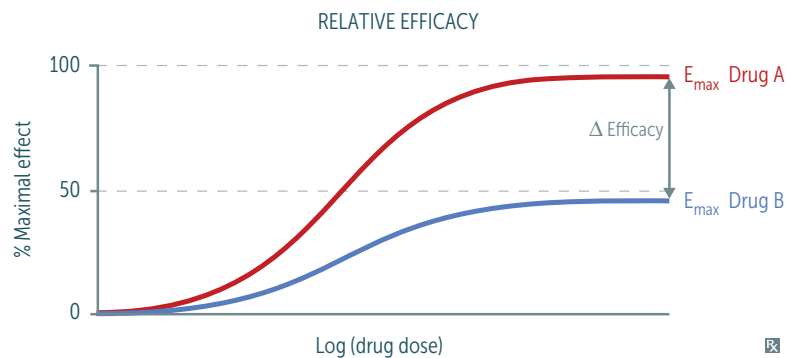
pKa

pH at which drugs (weak acid or base) are 50% ionized and 50% nonionized. The pKa represents the strength of the weak acid or base.

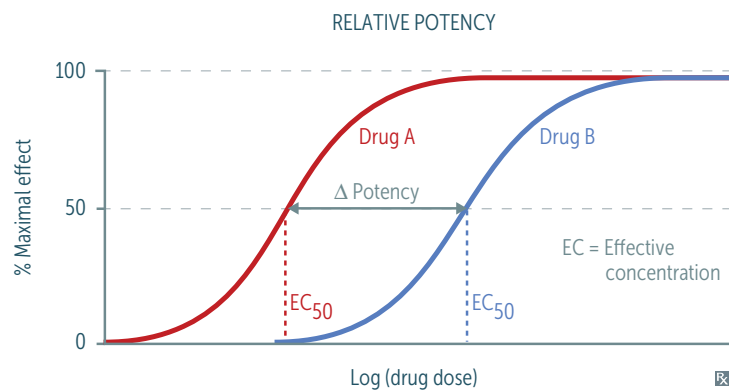


Efficacy vs potency**Efficacy**

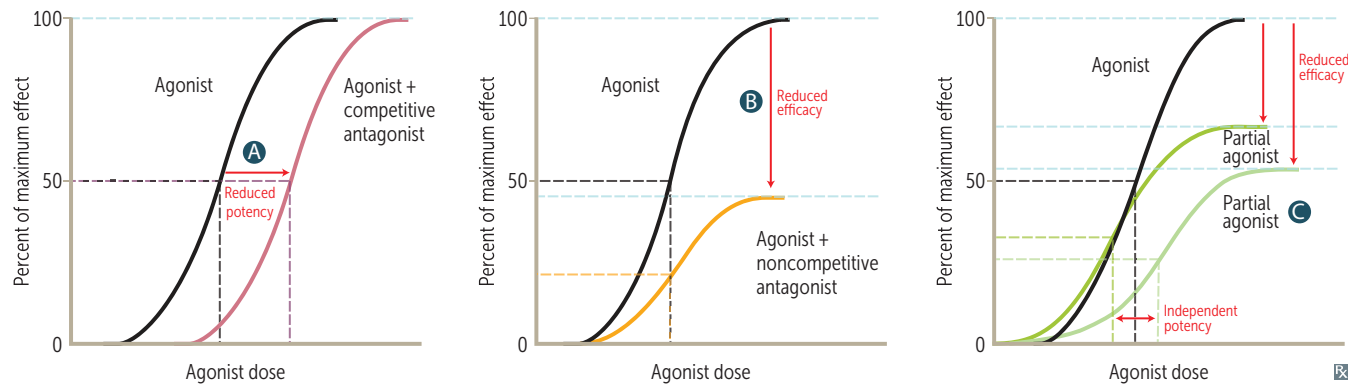
Maximal effect a drug can produce (intrinsic activity). Represented by the y-value ($E_{\max}$). $\uparrow$ y-value = $\uparrow E_{\max}$ = $\uparrow$ efficacy. Unrelated to potency (ie, efficacious drugs can have high or low potency). Partial agonists have less efficacy than full agonists.

**Potency**

Amount of drug needed for a given effect. Represented by the x-value (EC_{50}). Left shifting = $\downarrow EC_{50}$ = $\uparrow$ potency = $\downarrow$ drug needed. Unrelated to efficacy (ie, potent drugs can have high or low efficacy).



Receptor binding



AGONIST WITH	POTENCY	EFFICACY	REMARKS	EXAMPLE
A Competitive antagonist	↓	No change	Can be overcome by ↑ agonist concentration	Diazepam (agonist) + flumazenil (competitive antagonist) on GABA _A receptor.
B Noncompetitive antagonist	No change	↓	Cannot be overcome by ↑ agonist concentration	Norepinephrine (agonist) + phenoxybenzamine (noncompetitive antagonist) on α-receptors.
C Partial agonist (alone)	Independent	↓	Acts at same site as full agonist	Morphine (full agonist) vs buprenorphine (partial agonist) at opioid μ-receptors.
Inverse agonist (alone)	Independent	Independent	Binds to a constitutively active receptor, thereby reducing its activity; has the opposite effect of an agonist	H ₁ antihistamines (eg, diphenhydramine)

Therapeutic index

Measurement of drug safety.

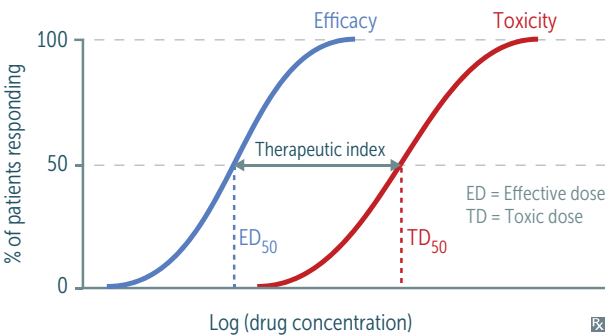
$$\frac{TD_{50}}{ED_{50}} = \frac{\text{median toxic dose}}{\text{median effective dose}}$$

Therapeutic window—range of drug concentrations that can safely and effectively treat disease.

TITE: Therapeutic Index = TD_{50} / ED_{50} .

Safer drugs have higher TI values. Drugs with lower TI values frequently require monitoring (eg, warfarin, theophylline, digoxin, antiepileptic drugs, lithium; **Warning! These drugs are lethal!**).

LD_{50} (lethal median dose) often replaces TD_{50} in animal studies.

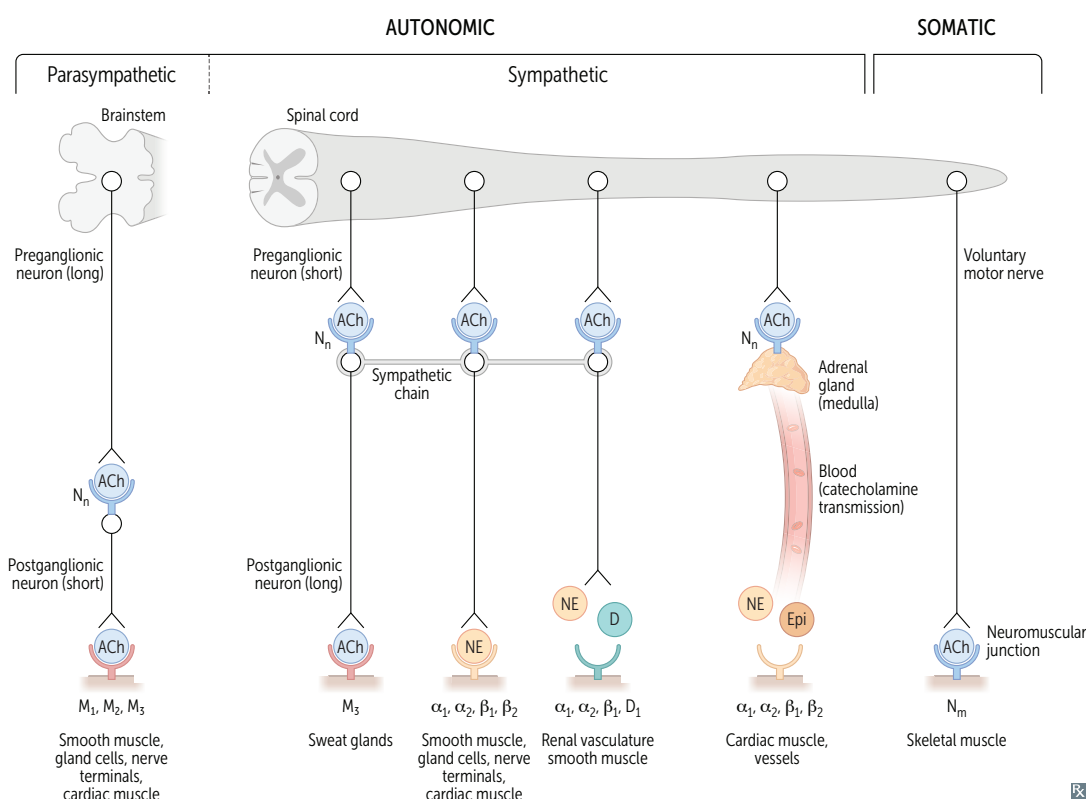


Drug effect modifications

TERM	DEFINITION	EXAMPLE
Additive	Effect of substances A and B together is equal to the sum of their individual effects	Aspirin and acetaminophen “2 + 2 = 4”
Permissive	Presence of substance A is required for the full effects of substance B	Cortisol on catecholamine responsiveness
Synergistic	Effect of substances A and B together is greater than the sum of their individual effects	Clopidogrel with aspirin “2 + 2 > 4”
Potentialiation	Similar to synergism, but drug B (with no therapeutic action alone) enhances the therapeutic action of drug A	Carbidopa only blocks enzyme to prevent peripheral conversion of levodopa “2 + 0 > 2”
Antagonistic	Effect of substances A and B together is less than the sum of their individual effects	Morphine with naloxone “2 + 2 < 4”
Tachyphylactic	Acute decrease in response to a drug after initial/repeated administration	Repeat use of intranasal decongestant (eg, oxymetazoline) → ↓ therapeutic response (with rebound congestion)

► PHARMACOLOGY—AUTONOMIC DRUGS

Autonomic receptors



Pelvic splanchnic nerves and CNs III, VII, IX and X are part of the parasympathetic nervous system. Adrenal medulla is directly innervated by preganglionic sympathetic fibers.

Sweat glands are part of the **sympathetic** pathway but are innervated by **cholinergic** fibers (**sympathetic** nervous system results in a “**chold**” sweat).